### Problem 9

# Problem 9

**Problem 1** *Find the derivatives of the following functions:*

(a) $f(x) = \sec^{-1}(\sqrt{x})$.

**Explanation.** $f'(x) = \dfrac{1}{\sqrt{x}\sqrt{x-1}} \cdot \dfrac{1}{2} x^{-\frac{1}{2}} = \dfrac{1}{2x\sqrt{x-1}}$

(b) $g(x) = \ln(\sin^{-1}(x))$.

**Explanation.** $g'(x) = \dfrac{1}{\sin^{-1}(x)} \cdot \dfrac{1}{\sqrt{1-x^2}}$.

(c) $h(x) = \dfrac{1}{\tan^{-1}(x^2+4)}$.

**Explanation.** $h'(x) = -\left(\tan^{-1}(x^2+4)\right)^{-2} \cdot \dfrac{1}{1+(x^2+4)^2} \cdot (2x)$.